

2 - AI Based Travel Planner Using Knowledge Bases

Introduction

Travel planning involves selecting destinations, activities, food options, and budgeting while considering the preferences of the traveler. Artificial Intelligence can assist in this process by using existing domain knowledge and applying reasoning techniques to generate personalized travel recommendations.

This project implements an AI-based Travel Planner that utilizes knowledge bases containing information about tourist attractions, food recommendations, and travel budgets. Based on user preferences, the system generates a personalized itinerary and estimates the overall trip cost.

Objective

The objectives of this project are:

- To design an AI-based Travel Planner.
- To reuse existing knowledge bases in the travel domain.
- To provide tourist place recommendations based on user interests.
- To recommend local food specialties.
- To generate personalized travel itineraries.
- To estimate trip costs based on budget preferences.

System Overview

The Travel Planner consists of the following components:

Knowledge Base

The knowledge base stores information related to:

- Tourist attractions
- Food recommendations
- Budget categories

The information is organized by destination and category, allowing efficient retrieval of relevant recommendations.

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Recommendation Engine

The recommendation engine analyzes user preferences and retrieves suitable tourist attractions and food recommendations from the knowledge base.

Inputs considered:

- Destination
- Area of interest (Beaches, History, Food, etc.)

The system returns relevant recommendations based on the selected category.

Itinerary Generator

The itinerary generator creates a day-wise travel plan using the recommendations obtained from the knowledge base.

The generated plan distributes attractions and food suggestions across the specified number of travel days.

Cost Assessment Module

The cost assessment module estimates the total trip cost using predefined budget categories:

- Low Budget
- Medium Budget
- High Budget

The estimated cost is calculated based on the selected budget level and trip duration.

Knowledge Representation

The system uses a rule-based knowledge representation approach.

Example:

Destination: Goa

Tourist Attractions:

- Baga Beach
- Calangute Beach
- Anjuna Beach
- Fort Aguada

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Food Recommendations:

- Fish Curry Rice
- Prawn Balchao
- Bebinca

Budget Categories:

- Low: ₹2000 per day
- Medium: ₹5000 per day
- High: ₹10000 per day

The planner retrieves and combines this information to generate recommendations.

Project Structure

AI_Travel_Planner/

- knowledge_base/
 - places.py
 - food.py
 - budgets.py
- planner/
 - recommender.py
 - itinerary.py
- tests/
 - test_planner.py
- docs/
 - report.md
- main.py

Algorithm

Step 1: Accept User Inputs

The user provides:

- Destination
- Interest Category
- Number of Days
- Budget Preference

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Step 2: Retrieve Knowledge

The recommendation engine searches the knowledge base for matching tourist attractions and food options.

Step 3: Generate Itinerary

The itinerary generator distributes recommended activities across the travel days.

Step 4: Estimate Cost

The cost assessment module calculates:

Estimated Cost = Daily Budget $\times$ Number of Days

Step 5: Display Results

The personalized itinerary and estimated cost are displayed to the user.

Test Case

Input

Destination: Goa

Interest: Beaches

Number of Days: 3

Budget: Medium

Expected Output

Day 1:

- Baga Beach
- Fish Curry Rice

Day 2:

- Calangute Beach
- Prawn Balchao

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Day 3:

- Anjuna Beach
- Bebinca

Estimated Cost: ₹15000

Results

The Travel Planner successfully generated personalized travel recommendations using the knowledge base.

The system was able to:

- Recommend tourist attractions based on interests.
- Suggest local food specialties.
- Create day-wise itineraries.
- Estimate trip costs according to user-selected budget levels.

The generated plans were relevant to user preferences and demonstrated the practical application of AI-based recommendation systems.

Output

```
Enter destination: Goa
Interest (Beaches/History/Food): Beaches
Number of days: 3
Budget (Low/Medium/High): Medium

PERSONALIZED TRAVEL PLAN

Day 1
- Baga Beach
- Try Fish Curry Rice

Day 2
- Calangute Beach
- Try Prawn Balchao

Day 3
- Anjuna Beach
- Try Bebinca

Estimated Cost: ₹15000
```

Conclusion

This project successfully demonstrates an AI-based Travel Planner that utilizes domain knowledge to generate personalized travel recommendations. By combining tourist attractions, food suggestions, and budget estimation, the system assists users in planning trips efficiently. The project illustrates how knowledge-based AI systems can be applied to solve practical travel planning problems.